

Homework 3: Qualitative Analysis II — The Chemistry

Paper track. Pairs with Home Lab 3. Do this on your own.

Part A: Formula recall (1 pt each)

Write the formula from memory:

1. Baking soda
2. Washing soda
3. Calcium carbonate
4. Sucrose
5. Starch
6. Table salt
7. Gypsum
8. Epsom salt
9. Citric acid
10. Vitamin C
11. Sodium acetate
12. Sand

Part B: Why the tests work (short answer)

1. **(4 pts)** Explain why and what makes corn starch turn black in iodine. (*Aim for four scoring points: the helix, the iodine going inside, the charge-transfer complex, the light absorption / color.*)
2. **(2 pts)** Is the iodine–starch test a chemical reaction? What is the correct term for what happens?
3. **(2 pts)** Amylose gives a blue-black color; amylopectin gives red-brown. Why the difference?
4. **(2 pts)** A powder turns iodine **colorless**. What is it, and why does that happen? How is this different from the starch result?
5. **(2 pts)** Why does chalk fizz more slowly with acid than baking soda does, even though both are carbonates?

Part C: Distinguish (2 pts each)

For each pair, name the single best test to tell them apart and the result for each:

1. Baking soda vs. baking powder
2. Table salt vs. citric acid
3. Corn starch vs. calcium carbonate
4. Granulated sugar vs. table salt

Part D: Benedict's (2 pts)

A question says "Benedict's solution turned brick-red." What does that mean? Would you expect table sugar to do this?

